

Multiple Choice

1. **Answer:** D

Options: A) Error; B) a; C) b; D) c

Note: In Python, negative indexing allows access to elements from the end of a sequence, so "abc"[-1] retrieves the last character, which is 'c'.

2. **Answer:** C

Options: A) 4; B) 1; C) 10; D) 8

Note: The correct answer is 10 because the sum function in Python 3 calculates the total of the elements in the list l, which are 1, 2, 3, and 4, resulting in $1 + 2 + 3 + 4 = 10$.

3. **Answer:** D

Options: A) The wait is expected to be short.; B) A busy-wait loop is easier to code than an interrupt handler.; C) There is no other work for the processor to do.; D) The program executes on a time-sharing system.

Note: The program executing on a time-sharing system indicates that resources are shared among multiple processes, making busy-waiting inefficient as it can waste CPU cycles while other processes could use the CPU more effectively.

4. **Answer:** D

Options: A) All bit strings whose number of zeros is a multiple of five; B) All bit strings starting with a zero and ending with a one; C) All bit strings with an even number of zeros; D) All bit strings with more ones than zeros

Note: The set of all bit strings with more ones than zeros cannot be described by a regular expression because it requires counting and comparing the number of ones and zeros, which exceeds the capabilities of regular languages that can only recognize patterns without maintaining such counts.

5. **Answer:** A

Options: A) 4; B) 1; C) 256; D) 16

Note: In Python, the expression $4*1**4$ evaluates as 4 multiplied by 1 raised to the power of 4, where the exponentiation operator (**) has higher precedence than multiplication (*), resulting in $4*1$, which equals 4.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the correct formula for the number of distinct

functions mapping a set A with m elements into a set B with n elements is n^m , not $n!/(n - m)!$.

7. **Answer:** True

Options: A) True; B) False

Note: $O(n^2)$ grows faster than $O(n)$ and $O(\log n)$ as the input size increases because the quadratic term dominates the linear and logarithmic terms, leading to a more rapid increase in function value as n becomes large.

8. **Answer:** False

Options: A) True; B) False

Note: Local caching of files can lead to inconsistencies because changes made to a file on one machine may not be immediately reflected on other machines until the cache is synchronized, resulting in potential discrepancies across the file system.

9. **Answer:** False

Options: A) True; B) False

Note: Priority queuing can lead to starvation for lower-priority jobs because higher-priority jobs may continuously receive service, preventing equal access to resources over time.

10. **Answer:** False

Options: A) True; B) False

Note: The function `random()` in Python 3 generates random numbers based on a pseudo-random number generator, but it does not set the starting integer value; instead, the seed value is set using the `random.seed()` function.

Long Form Response

11. **Answer:** `def get_max_occurring_char(str 1): | return ch`

Note: The provided answer correctly defines a Python function intended to find the maximum occurring character in a string, as indicated by its naming and structure, although it lacks the complete implementation details necessary for full functionality.

12. **Answer:** `def mul_list(nums 1,nums 2): | return list(result)`

Note: correct because it illustrates the use of a lambda function within the map function to apply multiplication element-wise to two lists, returning the result as a new list.

End of Answer Key